

Skilled mathematician with strong experience in theoretical modeling, practical implementation, and data analysis. Leveraging all three to create innovative slot mechanics and tools to drive successful games.

Experience

Mathematician | IGT [Everi] | February 2024-Present

- Pioneering innovative game design by co-designing novel features, leveraging mathematical modeling techniques to test and prove feasibility early in the development cycle. Architecting core game structures and features, and pitching new game concepts and features which align with and expand upon current market standards.
- Producing clear, accessible math documentation for cross-discipline review, ensuring effective communication with non-technical stakeholders.
- Building custom computational tools, including a reusable framework for computing Markov chains in C# with live Excel interoperability. Helping to reduce reliance on simulations and empowering peers to adopt more rigorous theoretical approaches, enabling a push towards reels-first design and removing long-standing blockers to the iteration cycle.
- Developing reusable data analytics infrastructure within the in-house prototyping/simulation tool, including detailed statistics and session tracking. Complemented this with a python program which uses machine learning to analyze session data and benchmark against high performing games.
- Conducting detailed competitive research, understanding market trends and standards.
- Assisting in the development and testing of new standard feature frameworks in TypeScript, including optimizing output data for effective communication with the front-end developer.
- Providing meaningful feedback on various games throughout the development cycle and collaborating closely with other mathematicians, designers, artists, and developers.

Education

M.S. in Mathematics | University of Illinois Chicago | 2024

B.S. in Mathematics with Highest Distinction | Minor in Physics | University of Illinois Urbana-Champaign | 2022

Training

Internship Network in the Mathematical Sciences (INMAS) | 2023-2024

Applied Python packages including pandas, seaborn, numpy, and others to learn and implement techniques from data science, statistics, and machine learning to analyze large data sets. Strengthened ability to translate advanced mathematical concepts into practical insights, bridging the gap between academia and industry.

Other Experience

Instructor | University of Illinois Chicago, Department of Mathematics, Statistics, and Computer Science | Summer 2023

- Led a group of 40 incoming college freshmen through an intensive, 15 hour per week workshop in order to increase their mathematical ability and help them place into higher level courses, with a 98% success rate.

Teaching Assistant | UIC, Department of Mathematics, Statistics, and Computer Science | Fall 2022-Spring 2024

- Taught Finite Mathematics for Business, Precalculus, and College Algebra
- Led 3 sections of 20+ students each semester through weekly worksheets, presentations, and quizzes. Fostered an active learning environment during group work time and prepared short topical lectures for each class. Held office hours both for my students as well as any other students seeking help with mathematics. Graded exams and quizzes.

Grader | University of Illinois Urbana Champaign, Department of Mathematics | Fall 2021

- Graded homework and exams of over 70 students in Discrete Mathematics. Provided meaningful feedback in a timely manner. Held office hours to assist students with homework, exam preparation, and general understanding of the course material.

Skills

Programming: C#, TypeScript, Python

Languages: English (native), Greek (intermediate: B1)

Other: Microsoft Excel, Visual Studio, Confluence, Jira, Perforce, Agile; Strong communication, teamwork, and leadership skills